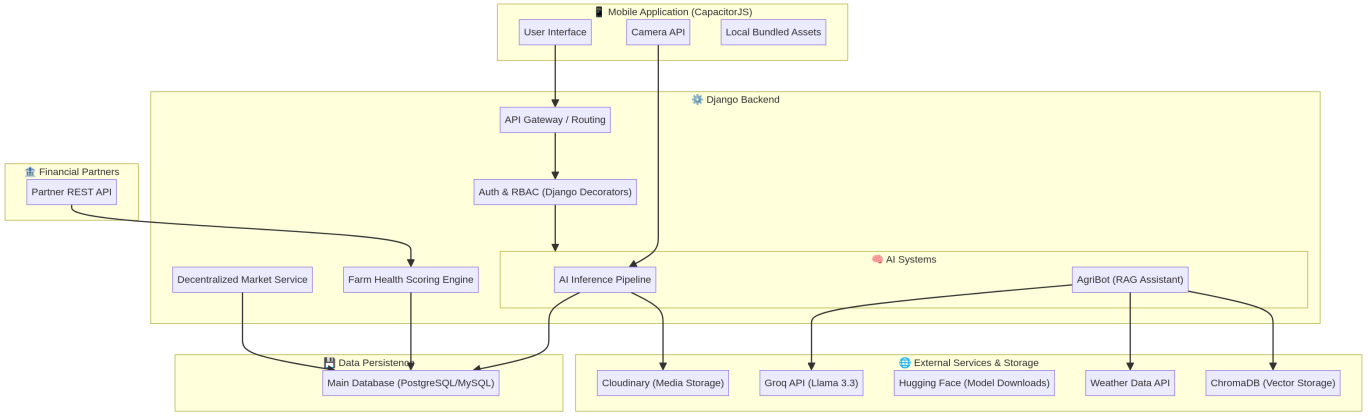
# AgriScan Technical Architecture Schematics

## Introduction

This document provides a detailed technical architecture overview of the AgriScan platform, outlining its core components, data flows, and interactions. AgriScan is designed with a decoupled, modular architecture to support real-time AI inference, complex data relationships, and robust financial integrations.

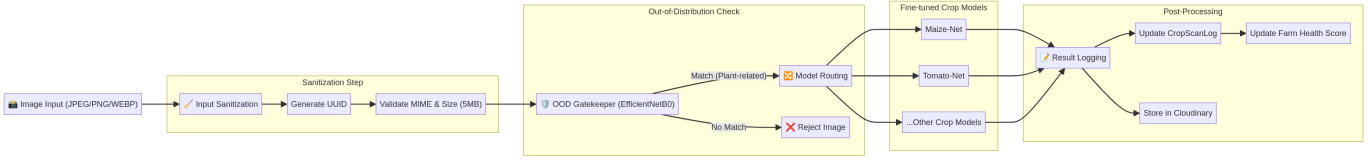
## 1. System Overview

This diagram illustrates the high-level architecture of the AgriScan platform, showcasing the primary components and their interconnections.




## 2. AI Inference Pipeline

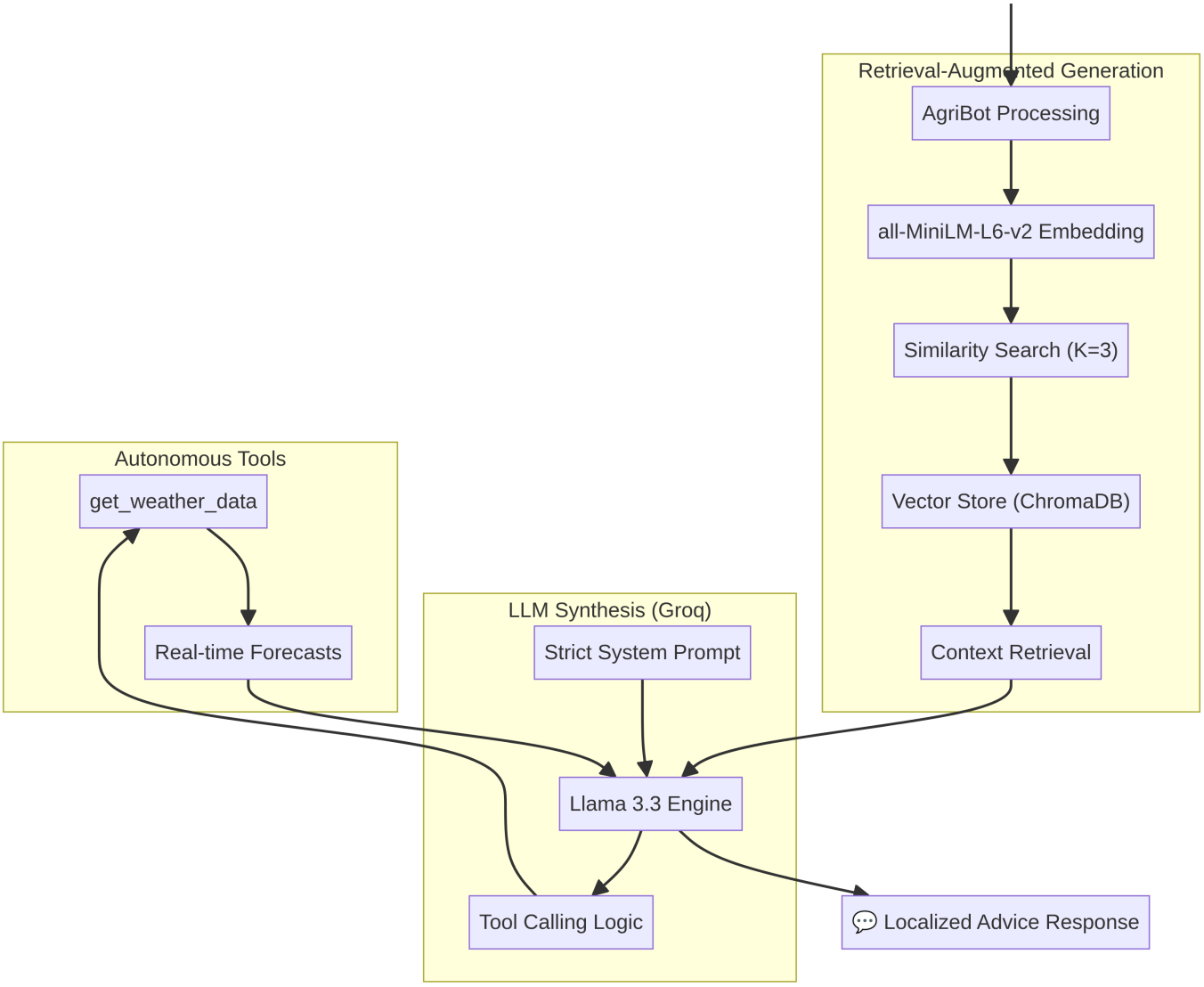
The AI Inference Pipeline is central to AgriScan's crop scanning capabilities, employing a multi-stage validation process for every image submitted.




## 3. RAG Assistant Architecture (AgriBot)

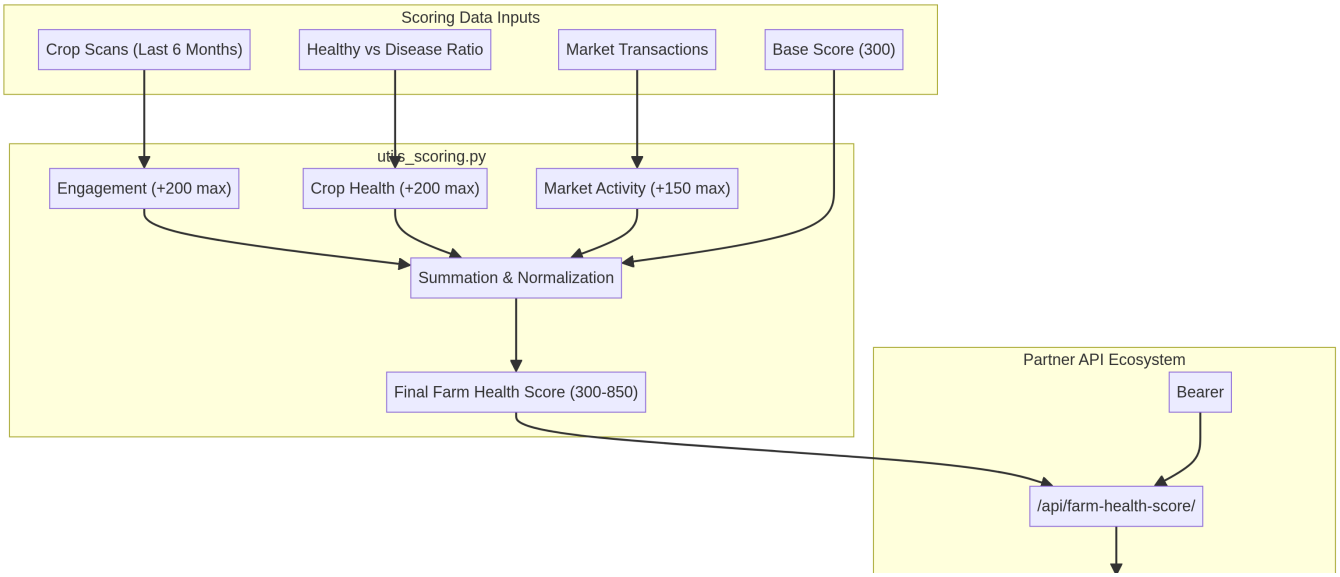
The AgriBot leverages Retrieval-Augmented Generation (RAG) to provide farmers with localized and context-aware advice.

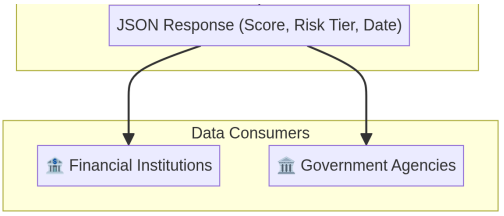





# 4. Financial Scoring & Partner API

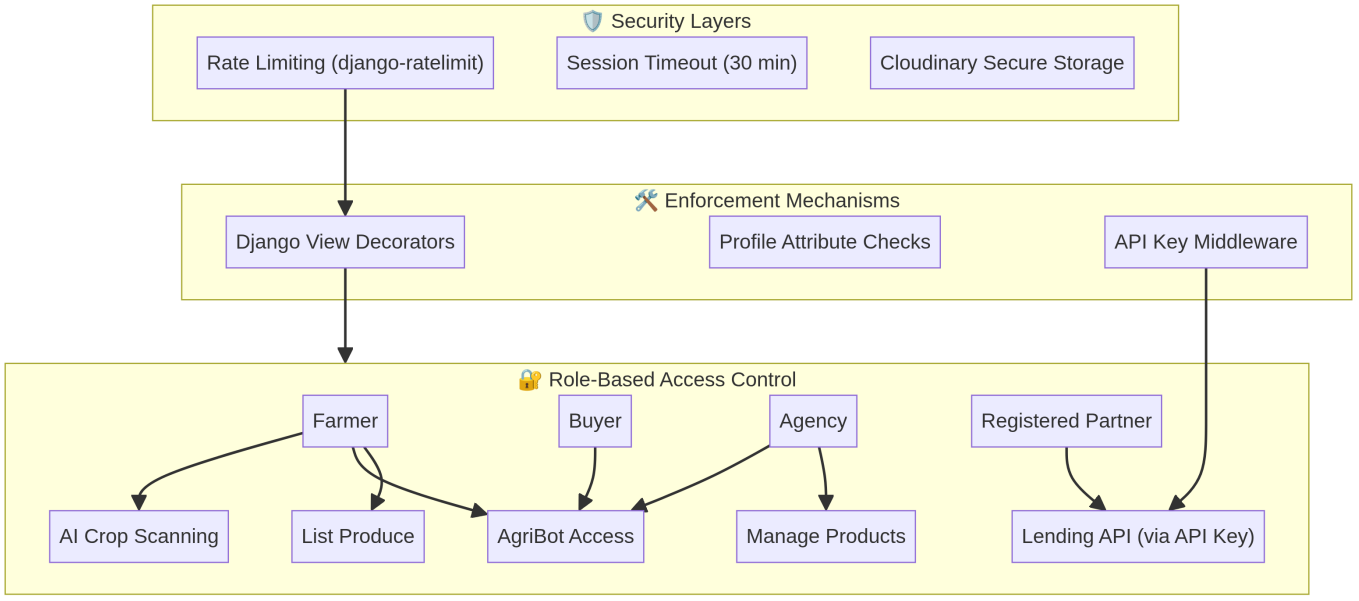
AgriScan transforms digital footprints into a verifiable creditworthiness metric, the Farm Health Score, which is accessible via a secure Partner API.





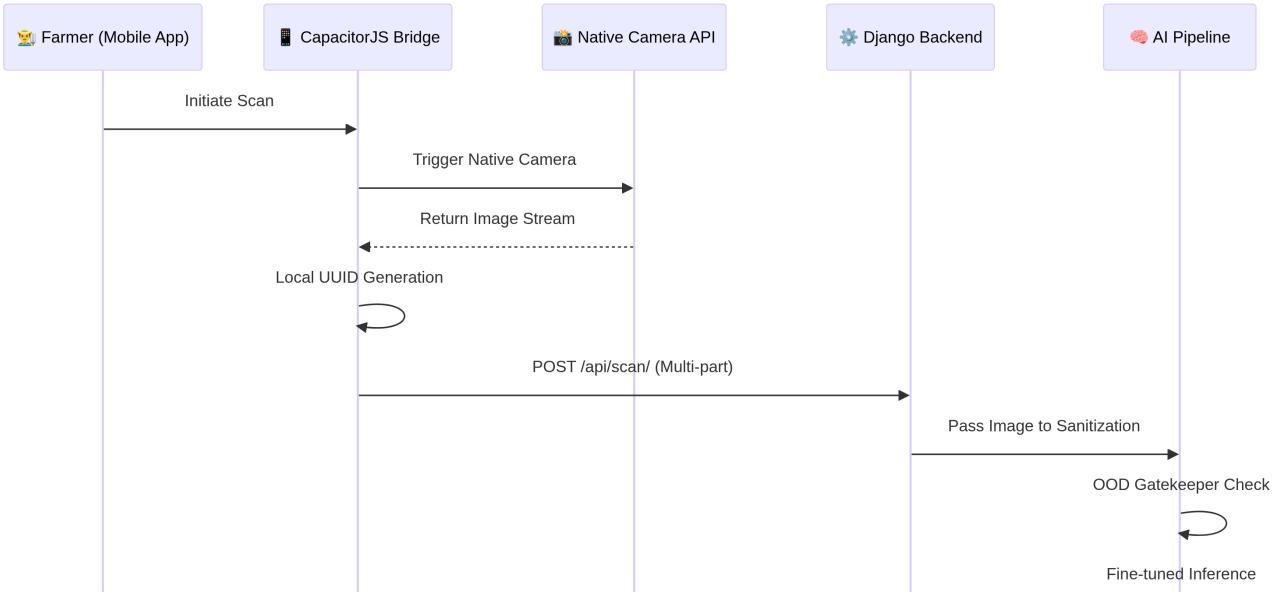

# 5. Security, RBAC, and Infrastructure

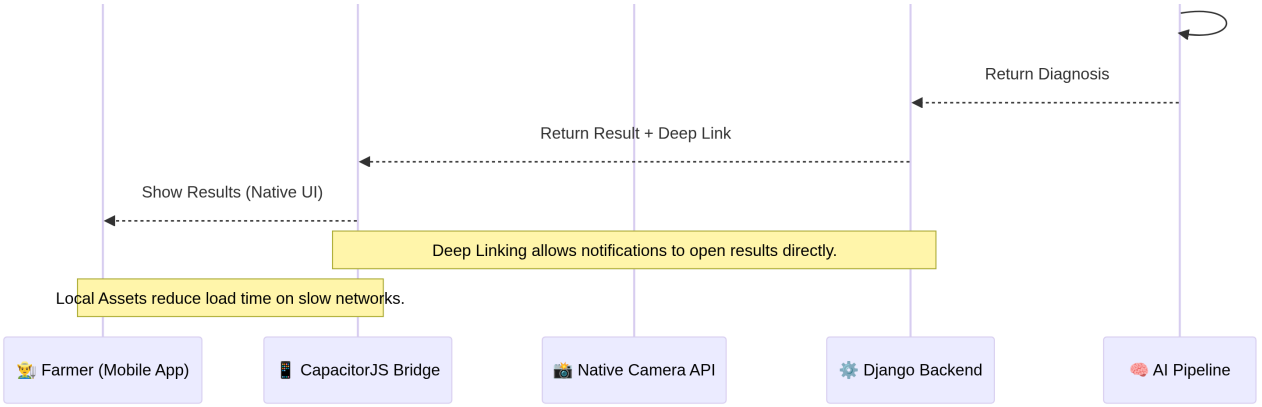
Security is paramount in AgriScan, implemented through robust Role-Based Access Control (RBAC) and infrastructure-level protections.




# 6. Mobile Native Integration

AgriScan is delivered as a native mobile application via CapacitorJS, ensuring a seamless and performant user experience with key mobile integrations.




Return Diagnosis
Return Result + Deep Link
Show Results (Native UI)
Deep Linking allows notifications to open results directly.
Local Assets reduce load time on slow networks.
👨‍🌾 Farmer (Mobile App)
📱 CapacitorJS Bridge
📸 Native Camera API
⚙️ Django Backend
🧠 AI Pipeline